

Chapter 1

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 1: Introduction to STEM – Your Gateway to Innovation! Objective: Dive into the exciting world of STEM through hands-on activities, quick brainstorming, and creative challenges. This worksheet is designed to spark curiosity and help high school students see STEM in action in their lives, careers, and the world around them—less writing, more doing!

Part 1: Quick STEM Spotter (15 points)

STEM is everywhere! Take 10 minutes to look around your home, school, or neighborhood (or think about your day). List 3 things you use or see that involve STEM (like a phone, a bike, or a water bottle). For each, jot down 1-2 words about which STEM area (Science, Technology, Engineering, Math) is behind it. Quick and easy!

STEM Thing	STEM Area(s) Involved
Example: Smartphone	Technology, Engineering
• 1.	
• 2.	
• 3.	

STEM Thing

STEM Area(s) Involved

Example: Smartphone

Technology, Engineering

- 1.

- 2.
- 3.

Part 2: STEM Dream Job Brainstorm (15 points)

Imagine your dream STEM job from the chapter (like game developer, marine biologist, or rocket engineer). Answer these quick prompts with short phrases or bullet points—no long sentences needed! Then, doodle a small sketch of yourself in that role (stick figures are fine!).

- **My Dream STEM Job:** _____
- **Why It’s Cool:** (1-2 reasons)
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- **One Skill I’d Need:** _____
- **How I Could Start Learning It Now:** (1 idea, like a club or app)
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Sketch Space: Draw yourself in your STEM dream job! (e.g., coding at a desk, exploring the ocean, building a robot) [Blank Space Here for Doodling]

Part 3: STEM Explorer Challenge – Pick Your Adventure! (30 points)

Choose one of these fun, hands-on challenges to bring STEM to life. These aren’t about writing essays—they’re about experimenting and thinking like a STEM innovator. Pick what excites you!

Option A: STEM Scavenger Hunt Go on a mini scavenger hunt (at home or school) to find 5 items or examples of STEM in action. Snap a quick photo of each (or describe them if no camera). For each, note one way it connects to Science, Technology, Engineering, or Math. Use the table below to track your finds!

Item/Example	Where I Found It	STEM Connection (1 way)
Example: Microwave	Kitchen	Technology – heats food with waves
• 1.		
• 2.		
• 3.		
• 4.		

• 5.		

Item/Example

Where I Found It

STEM Connection (1 way)

Example: Microwave

Kitchen

Technology – heats food with waves

- 1.
- 2.
- 3.
- 4.
- 5.

Option B: Build a STEM Gizmo Grab household stuff (straws, tape, paper, spoons—whatever you’ve got!) and build a mini gizmo or structure, like a tiny bridge, tower, or boat. Test it out (e.g., can it hold a small toy? Float in water?). Then, answer these quick prompts:

- **What I Built:** _____
- **Did It Work? (Yes/No & Why):** _____
- **One STEM Idea I Used:** (e.g., balance, strength) _____ Attach a photo or quick sketch of your creation if you can! (Stick figures or rough lines are cool.) [Space Here for Sketch or Photo Reference]

Part 4: Big Picture STEM Impact (10 points)

STEM solves real-world problems! Pick one global challenge from the chapter (like climate change, healthcare, or food security). In just 2-3 quick bullet points, brainstorm one wild or cool way STEM could help solve it. Then, add one tiny action you could do to support this cause (like sharing a post or trying something at home). Keep it short and creative!

- **Challenge I Picked:** _____
- **Cool STEM Solution Idea:**
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 -
 -
- **My Tiny Action:** _____

Part 5: Quick Reflection – What’s Your STEM Spark? (10 points)

Think about Chapter 1 for a sec. Answer these in 1-2 short phrases each. What got you hyped about STEM? Where

do you see it in your future?

- **Coolest Thing I Learned About STEM:** _____
- **How I Might Use STEM Someday:** (hobby, job, or just for fun)
